Cyber Security Course

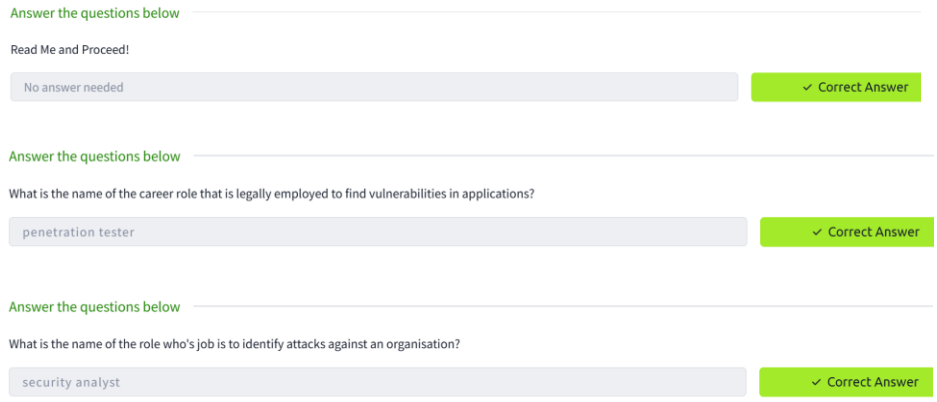
Submitted By: Maryam Rehan

Mentor: Sir Abdullah

Assignment : 03

TRY HACK ME ROOM





Task 1: Welcome to TryHackMe

Objective: To understand the purpose of TryHackMe and explore different cybersecurity career paths.

Observation: I learned that TryHackMe provides practical cybersecurity learning through guided rooms, challenges, and learning paths.

Result: Completed – Read Me and Proceed.

Task 2: Offensive Security

Objective: To understand offensive security and the role of penetration testers.

Observation: Offensive Security focuses on finding vulnerabilities in applications and systems. A Penetration Tester is legally employed to identify security weaknesses.

Result: Completed – Identified the role of a Penetration Tester.

Task 3: Defensive Security

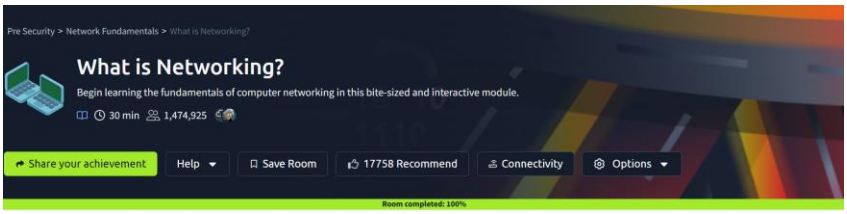
Objective: To understand defensive security and its career roles.

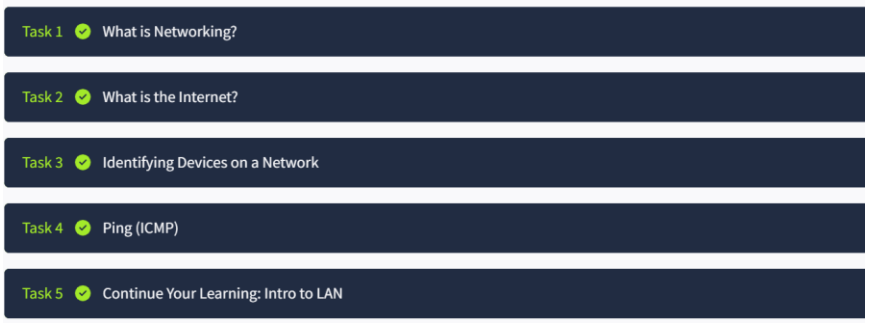
Observation: Defensive Security focuses on detecting and preventing cyberattacks. I learned about Security Analysts, Incident Responders, and Malware Analysts.

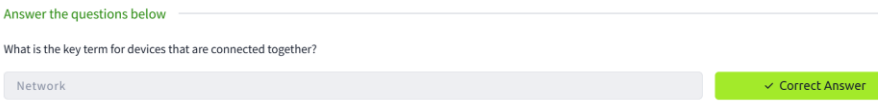
Result: Completed – Identified Security Analyst as the role responsible for detecting attacks against an organisation.

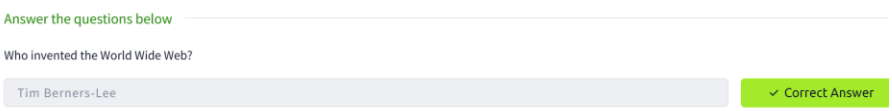
Conclusion

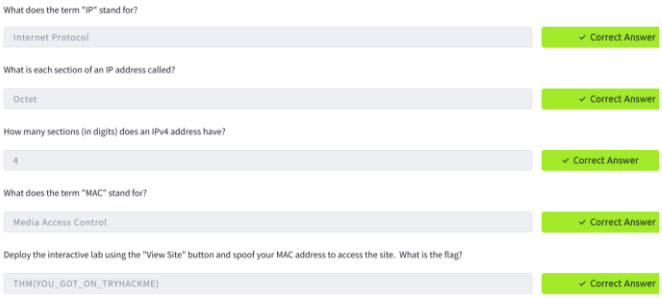
This room helped me understand the basic difference between Offensive and Defensive Security and introduced me to different cybersecurity career opportunities.













Task 1: What is Networking?

Objective: To understand the basic concept of computer networking.

Observation: Networking is the process of connecting computers and devices to share information and resources.

Result: Learned the basic concept of networking.

Task 2: What is the Internet?

Objective: To understand what the Internet is and how it works.

Observation: The Internet is a global network that connects millions of computers and devices to communicate and exchange information.

Result: Understood the basic concept of the Internet.

Task 3: Identifying Devices on a Network

Objective: To learn about different devices used in a network.

Observation: Devices such as routers, switches, computers, and servers help devices communicate within a network.

Result: Learned about common network devices and their functions.

Task 4: Ping (ICMP)

Objective: To understand the purpose of Ping and ICMP.

Observation: Ping is a network utility used to check connectivity between devices. It uses ICMP messages to test whether a destination is reachable.

Result: Learned how Ping is used for basic network connectivity testing.

Task 5: Continue Your Learning – Intro to LAN

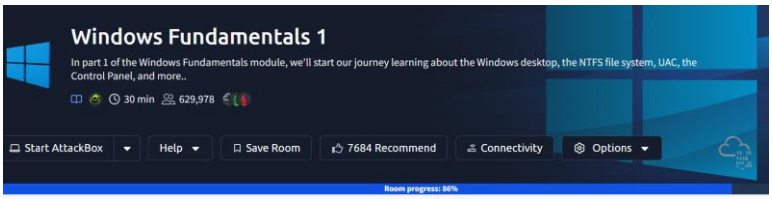
Objective: To explore the fundamentals of Local Area Networks (LAN).

Observation: A LAN connects devices within a limited geographical area, such as a home, school, or office.

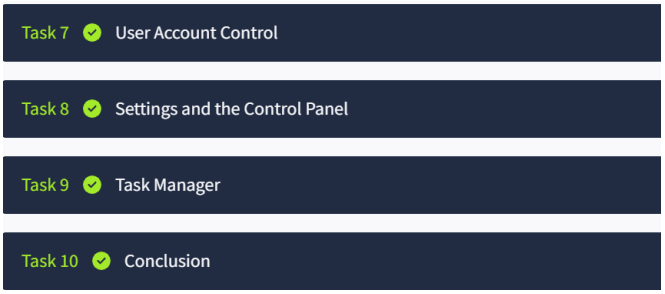
Result: Learned the basic concept and purpose of a LAN.

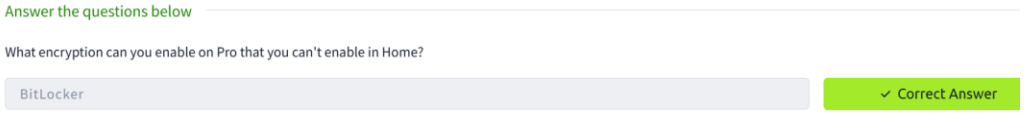
Conclusion

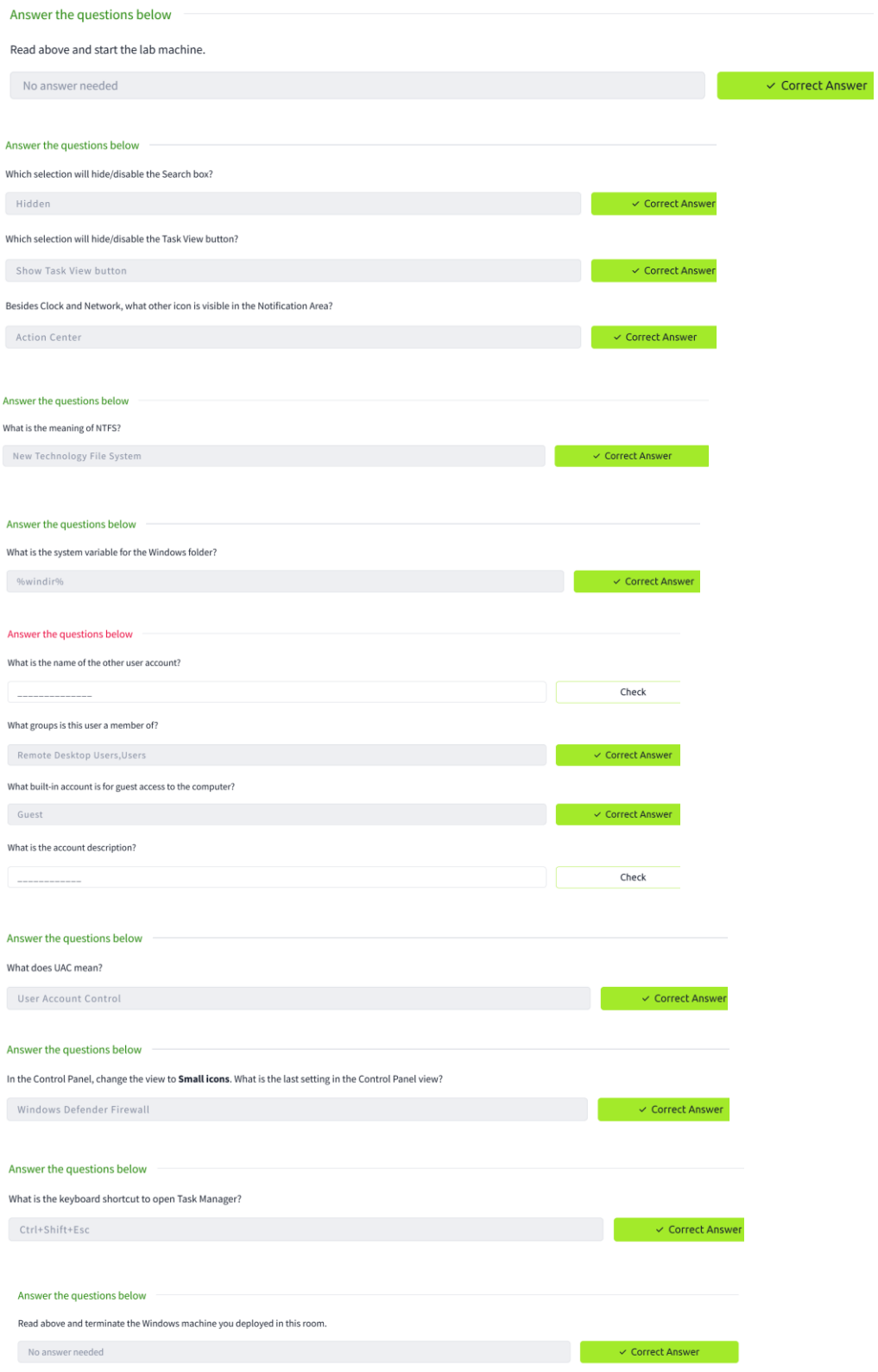
This room helped me understand networking, the Internet, network devices, Ping (ICMP), and Local Area Networks. It improved my understanding of basic network communication.











Task 1: Windows Editions

Objective: To understand different editions of Windows.

Observation: Windows has different editions such as Home, Pro, and Enterprise. Each edition provides different features.

Result: Learned about Windows editions.

Task 2: Introduction to Windows

Objective: To understand the Windows operating system.

Observation: Windows is an operating system that manages computer hardware, software, files, and applications.

Result: Learned the basic functions of Windows.

Task 3: The Desktop (GUI)

Objective: To understand the Windows graphical user interface.

Observation: The desktop provides access to icons, the Start Menu, taskbar, and other system features.

Result: Learned about the Windows desktop environment.

Task 4: The File System

Objective: To understand how Windows stores and organizes files.

Observation: Windows uses drives, folders, and files to organize and manage data.

Result: Learned basic file system concepts.

Task 5: The Windows\System32 Folders

Objective: To understand the purpose of the System32 folder.

Observation: System32 contains important Windows system files, libraries, and utilities required for system operations.

Result: Learned about the importance of the System32 folder.

Task 6: User Accounts, Profiles, and Permissions

Objective: To understand Windows user accounts and permissions.

Observation: This task could not be completed because of a technical error.

Result: Pending due to an error.

Task 7: User Account Control

Objective: To understand User Account Control (UAC).

Observation: UAC helps prevent unauthorized changes by requesting permission before certain administrative actions.

Result: Learned the purpose of UAC.

Task 8: Settings and the Control Panel

Objective: To explore Windows system configuration options.

Observation: Settings and Control Panel allow users to manage system preferences, devices, accounts, and other configurations.

Result: Learned basic Windows configuration options.

Task 9: Task Manager

Objective: To understand how Task Manager monitors system activity.

Observation: Task Manager displays running applications, processes, CPU, memory, and other resource usage.

Result: Learned how to monitor system performance.

Task 10: Conclusion

Objective: To review the concepts covered in the room.

Observation: Reviewed the main Windows features and system management tools.

Result: Completed the introductory review.

Overall Conclusion

This practical room helped me understand Windows editions, desktop environment, file management, system folders, user accounts, security controls, and system monitoring. I also learned how Windows Settings, Control Panel, and Task Manager help manage a computer.



Task 1 What is HTTP(S)?

Task 2 Requests And Responses

Task 3 HTTP Methods

Task 4 HTTP Status Codes

Task 5 Headers

Task 6 Cookies

Task 7 Making Requests

Answer the questions below

What does HTTP stand for?

HyperText Transfer Protocol ✓ Correct Answer

What does the S in HTTPS stand for?

secure ✓ Correct Answer

On the mock webpage on the right there is an issue, once you've found it, click on it. What is the challenge flag?

THM{INVALID_HTTP_CERT} ✓ Correct Answer

Answer the questions below

What HTTP protocol is being used in the above example?

HTTP/1.1 ✓ Correct Answer

What response header tells the browser how much data to expect?

Content-Length ✓ Correct Answer

Answer the questions below

What method would be used to create a new user account?

POST ✓ Correct Answer

What method would be used to update your email address?

PUT ✓ Correct Answer

What method would be used to remove a picture you've uploaded to your account?

DELETE ✓ Correct Answer

What method would be used to view a news article?

GET ✓ Correct Answer

Answer the questions below

What response code might you receive if you've created a new user or blog post article?

201 ✓ Correct Answer

What response code might you receive if you've tried to access a page that doesn't exist?

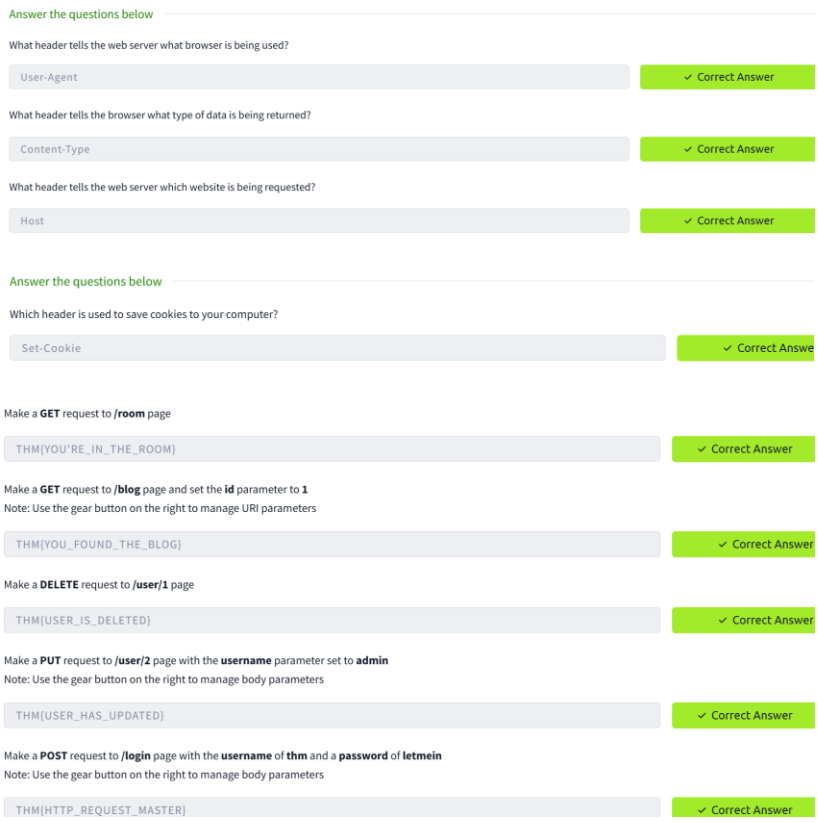
404 ✓ Correct Answer

What response code might you receive if the web server cannot access its database and the application crashes?

503 ✓ Correct Answer

What response code might you receive if you try to edit your profile without logging in first?

401 ✓ Correct Answer



1. Introduction

HTTP (HyperText Transfer Protocol) is a communication protocol used by web browsers and web servers to exchange information. HTTPS is the secure version of HTTP, which uses encryption to protect data during communication.

2. Task 1: What is HTTP(S)?

In this task, I learned about HTTP and HTTPS, their purpose, and the importance of secure communication between browsers and web servers. HTTPS protects information through encryption.

3. Task 2: Requests and Responses

I learned how a browser sends an HTTP request to a web server and receives an HTTP response. A response contains a status code, headers, and sometimes webpage content.

4. Task 3: HTTP Methods

I learned about the four common HTTP methods:

- **GET:** Retrieves information from a server.
- **POST:** Sends data to a server.

- **PUT:** Updates existing information.
- **DELETE:** Removes information.

5. Task 4: HTTP Status Codes

HTTP status codes indicate the result of a request.

- **200 OK:** Request successful.
- **201 Created:** Resource created successfully.
- **401 Unauthorized:** Authentication is required.
- **404 Not Found:** Requested resource was not found.
- **500 Internal Server Error:** Server encountered an error.
- **503 Service Unavailable:** Server is temporarily unavailable.

6. Task 5: Headers

I learned that HTTP headers provide additional information about requests and responses.

Examples:

- **Host:** Specifies the requested website.
- **User-Agent:** Identifies the browser software.
- **Content-Type:** Describes the type of data being sent or received.
- **Content-Length:** Specifies the size of the message body.

7. Task 6: Cookies

Cookies are small pieces of data stored by a browser. Websites use cookies to maintain sessions and remember certain user information. The Set-Cookie header is used by a server to send cookies to a browser.

8. Task 7: Making Requests

This task introduced practical HTTP requests using the TryHackMe browser emulator. The activities involved:

1. Sending a GET request to /room.
2. Sending a GET request to /blog with the URI parameter id=1.
3. Sending a DELETE request to /user/1.
4. Sending a PUT request to /user/2 with the body parameter username=admin.

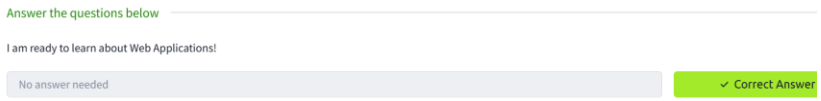
5. Sending a POST request to /login with username thm and password letmein.

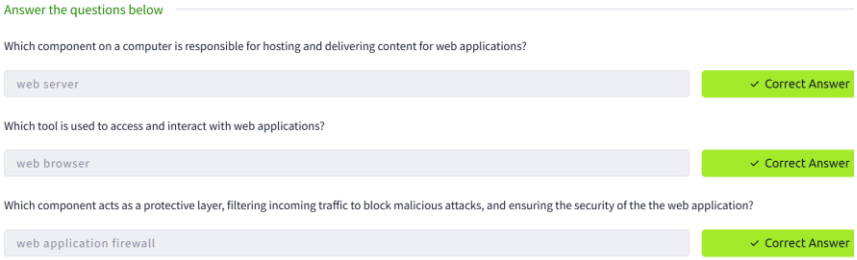
9. Conclusion

This room helped me understand how HTTP and HTTPS work, how web browsers communicate with servers, and how HTTP methods, status codes, headers, and cookies are used. I also learned the basic process of making HTTP requests through a browser emulator. These concepts are useful for understanding web communication and basic web security.









Answer the questions below

Which protocol provides encrypted communication to ensure secure data transmission between a web browser and a web server?

HTTPS ✓ Correct Answer

What term describes the practice of registering domain names that are misspelt variations of popular websites to exploit user errors and potentially engage in fraudulent activities?

Typosquatting ✓ Correct Answer

What part of a URL is used to pass additional information, such as search terms or form inputs, to the web server?

Query String ✓ Correct Answer

Answer the questions below

Which HTTP message is returned by the web server after processing a client's request?

HTTP response ✓ Correct Answer

What follows the headers in an HTTP message?

Empty Line ✓ Correct Answer

Answer the questions below

Which HTTP protocol version became widely adopted and remains the most commonly used version for web communication, known for introducing features like persistent connections and chunked transfer encoding?

HTTP/1.1 ✓ Correct Answer

Which HTTP request method describes the communication options for the target resource, allowing clients to determine which HTTP methods are supported by the web server?

OPTIONS ✓ Correct Answer

In an HTTP request, which component specifies the specific resource or endpoint on the web server that the client is requesting, typically appearing after the domain name in the URL?

URL Path ✓ Correct Answer

Answer the questions below

Which HTTP request header specifies the domain name of the web server to which the request is being sent?

Host ✓ Correct Answer

What is the default content type for form submissions in an HTTP request where the data is encoded as key=value pairs in a query string format?

application/x-www-form-urlencoded ✓ Correct Answer

Which part of an HTTP request contains additional information like host, user agent, and content type, guiding how the web server should process the request?

Request Headers ✓ Correct Answer

Answer the questions below

What part of an HTTP response provides the HTTP version, status code, and a brief explanation of the response's outcome?

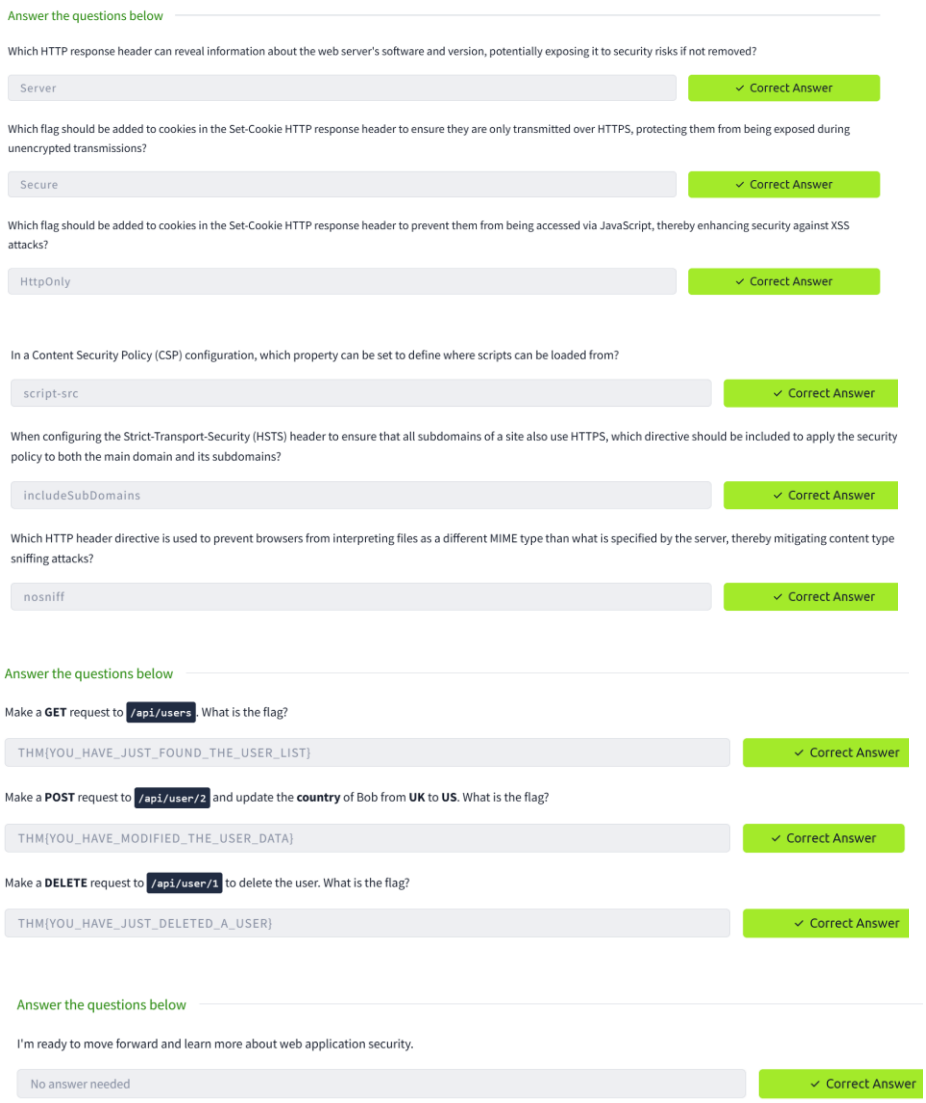
Status Line ✓ Correct Answer

Which category of HTTP response codes indicates that the web server encountered an internal issue or is unable to fulfil the client's request?

Server Error Responses ✓ Correct Answer

Which HTTP status code indicates that the requested resource could not be found on the web server?

404 ✓ Correct Answer



1. Introduction

Web applications are software programs that run through web browsers. They allow users to access online services, interact with websites, and exchange information over the internet. This room introduced the basic components and communication methods used in web applications.

2. Task 2: Web Application Overview

I learned about the main components of a web application:

- **Web Server:** Hosts and delivers content for web applications.
- **Web Browser:** Used to access and interact with web applications.
- **Web Application Firewall (WAF):** Filters incoming traffic and helps protect web applications from malicious attacks.

3. Task 3: Uniform Resource Locator (URL)

A URL is a web address used to locate online resources. Its main components include:

- **Scheme:** Defines the protocol, such as HTTP or HTTPS.
- **Host/Domain:** Identifies the website.
- **Port:** Directs communication to a specific service.
- **Path:** Identifies a particular page or resource.
- **Query String:** Passes additional information to the server.
- **Fragment:** Points to a specific section of a webpage.

I also learned about typosquatting, where attackers register misspelled versions of popular domain names to trick users.

4. Task 4: HTTP Messages

HTTP messages are exchanged between a client and a web server. There are two main types:

- **HTTP Request:** Sent by the client to the server.
- **HTTP Response:** Returned by the server after processing the request.

An HTTP message contains a start line, headers, an empty line, and a body.

5. Task 5: HTTP Request – Request Line and Methods

An HTTP request line contains the method, URL path, and HTTP version.

Common HTTP methods include:

- **GET:** Retrieves information.
- **POST:** Sends data to the server.
- **PUT:** Updates or replaces information.
- **DELETE:** Removes a resource.
- **OPTIONS:** Identifies the communication methods supported by a resource.

I also learned about HTTP versions, including HTTP/1.1, HTTP/2, and HTTP/3.

6. Task 6: HTTP Request – Headers and Body

Request headers provide additional information to the web server.

Examples include:

- **Host:** Specifies the requested domain.
- **User-Agent:** Identifies the browser.
- **Referer:** Shows the source URL.
- **Cookie:** Contains stored browser information.
- **Content-Type:** Specifies the format of the request data.

The request body contains data sent to the server. Common formats include URL Encoded, Form Data, JSON, and XML.

7. Task 7: HTTP Response – Status Line and Status Codes

The HTTP response status line contains the HTTP version, status code, and reason phrase.

Common status codes include:

- **200 OK:** Request successful.
- **301 Moved Permanently:** Resource has moved.
- **404 Not Found:** Requested resource does not exist.
- **500 Internal Server Error:** Server encountered an error.

HTTP status codes are divided into informational, successful, redirection, client error, and server error categories.

8. Task 8: HTTP Response – Headers and Body

Response headers provide information about the server response.

Examples include:

- **Date:** Shows when the response was generated.
- **Content-Type:** Identifies the response data format.
- **Server:** Identifies the server software.
- **Set-Cookie:** Sends cookies to the browser.
- **Cache-Control:** Controls response caching.
- **Location:** Specifies a redirection destination.

The response body contains the actual content, such as HTML, JSON, or images.

9. Task 9: Security Headers

I learned about important HTTP security headers that help protect web applications:

- **Content-Security-Policy (CSP):** Controls trusted sources for content and scripts.
- **Strict-Transport-Security (HSTS):** Enforces HTTPS connections.
- **X-Content-Type-Options:** Prevents MIME type sniffing using the nosniff directive.
- **Referrer-Policy:** Controls how much referrer information is shared.

These headers help reduce risks such as Cross-Site Scripting (XSS) and other web attacks.

10. Task 10: Practical Task – Making HTTP Requests

In this task, I learned to use an HTTP request emulator to interact with a demo web application.

The practical activities included:

1. Sending a GET request to /api/users to retrieve user information.
2. Sending a POST request to /api/user/2 to update Bob's country from UK to US.
3. Sending a DELETE request to /api/user/1 to remove a user.

This activity helped me understand how HTTP methods work in practical web application communication.

11. Task 11: Conclusion

This room improved my understanding of web application components, URLs, HTTP requests and responses, methods, headers, status codes, and security headers. The practical activity also provided an introduction to interacting with web applications through HTTP requests. These concepts are important for understanding web technologies and basic web application security.